CSE5026: Foundation and Frontiers of Cognitive Science

Memory (I)

Yang Xu

Department of Computer Science and Engineering

Southern University of Science and Technology

Overview

- Memory Systems
- Short-Term Memory
- Working Memory（工作记忆)
- Memory Mechanisms in AI/LLMs

Introduction and Brief History

获取/积累　　保存　　提取

- Memory is the mental process of acquiring, retaining, and retrieving individuals' experiences.

编码

- Using information-processing terms, memory involves **encoding**, **storing**, and **retrieving** information from the environment.

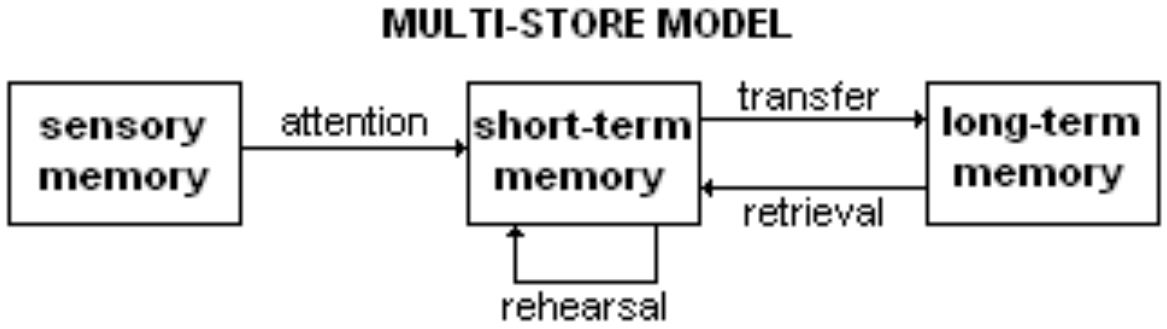
存储　　提取

- Memory enables mental activity the *continuity* over time.
  It connects successive experiences, allowing mental activity to develop as a unified process.

Models of Memory

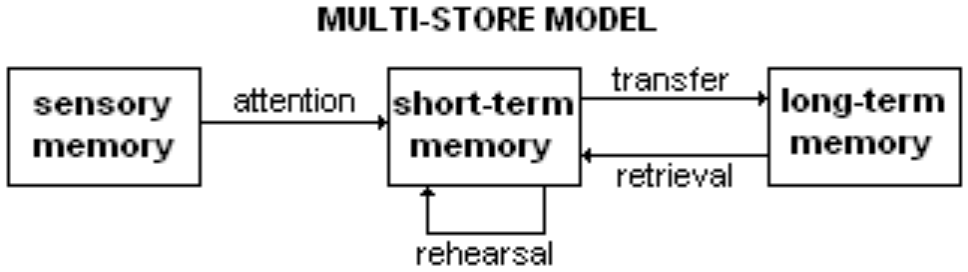
编码/获取　　存储/保存

- Human memory involves three stages: encoding (acquisition), storage (retention), and retrieval.　提取
- Atkinson-Shiffrin's **multi-store model** of memory (1968)
- (多存储模型)
- A simple linear model of human memory



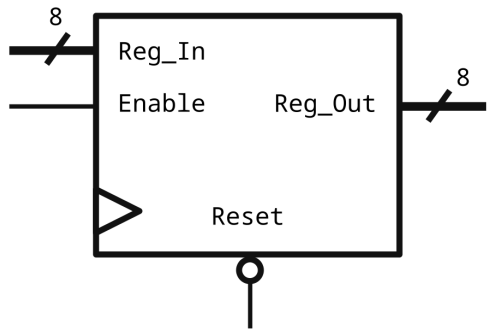
Source: https://en.wikipedia.org/wiki/Memory

The Multi-Store Model



- Atkinson–Shiffrin's multi-store model
- Basic assumption: memory consists of three distinct stores, which are conceptual components rather than physical organs.
- **Sensory register (SR)** (感觉寄存器)
- **Short-term store (STS)** (短时记忆存储)
- **Long-term store (LTS)** (长时记忆存储)

"register": analogy from digital circuit



Source: https://en.wikipedia.org/wiki/Atkinson–Shiffrin_memory_model - Sensory_register

The Sensory Register (SR)

感官

- When the senses detect a stimulus, it briefly holds that information in the **sensory register (SR)**, also called a **sensory buffer**, sensory memory, or instant memory.

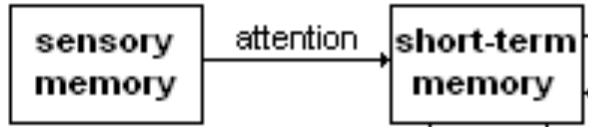
缓冲区

视 听

- Each sense has its own register. Most research focuses on vision and hearing.
- The registers prevent the flood of sensory input from overwhelming higher cognitive functions.

注意力

- *Attention* transfers selected information to short-term memory.



- Without attention, the information rapidly *decays* and is forgotten.

衰减

Source: https://en.wikipedia.org/wiki/Atkinson–Shiffrin_memory_model - Sensory_register

The Visual Register: a.k.a. Iconic Memory

映像记忆

- Iconic memory is associated with the visual system.
  - Example: Visual persistence (视觉残留). Moving a lit sparkler quickly produces a continuous trail of light rather than isolated points, showing that visual information does not disappear immediately.





Iconic Memory: Sperling's Experiment

- Sperling's (1960) partial-report experiment
- Participants briefly see a 3 × 3 array of nine letters for just 0.05 seconds (50 ms), far less time than needed to clearly identify each letter.
- When asked to report all the letters, most participants recall only 4–5.

R	N	P
A	B	D
E	Y	Q

shown for <50 ms

Source: https://doi.org/10.1037%2Fh0093759

Iconic Memory: Sperling's Experiment

- Sperling's (1960) partial-report experiment
- In partial report, a high, medium, or low **tone** (声音信号) immediately after the display cues participants to report row 1, 2, or 3, respectively.
- Participants report the three letters in any cued row with **nearly 100%** accuracy.
- Conclusion: All nine letters were actually briefly *available* in **iconic memory**. Information that did not receive attention quickly disappeared (decay).

短暂保存

R	N	P
A	B	D
E	Y	Q

shown for <50 ms

Source: https://doi.org/10.1037%2Fh0093759

The Auditory Register: a.k.a. Echoic Memory

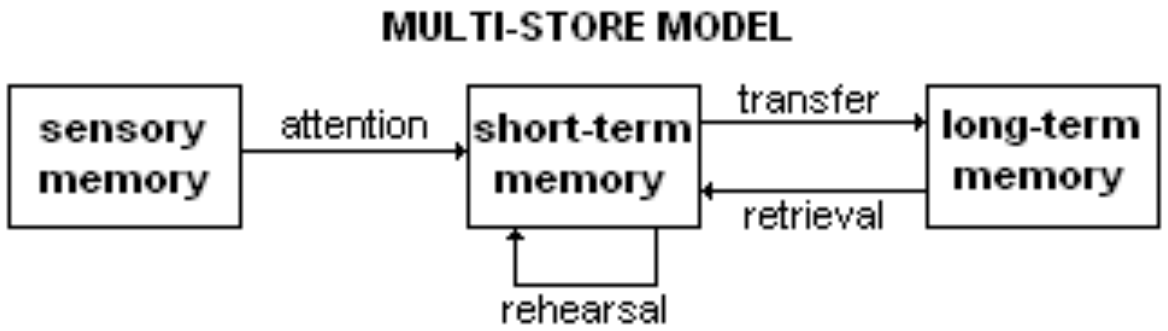
- Echoic memory briefly retains information in the auditory system.
  - Example: after music ends, its melody, rhythm, and instrumental sounds may continue to echo briefly in your mind.
- Duration: 1.5–5 seconds, extending to 20 seconds without interference. (余音绕梁)

Source: https://en.wikipedia.org/wiki/Atkinson–Shiffrin_memory_model - Echoic_memory

Levels of Memory

- Processing and retention of stimuli
  - Iconic memory (visual) lasts about **1 second**.
    Echoic memory (auditory) lasts **2–4 seconds**.
  - Short-term memory lasts seconds to **tens of seconds**.
  - Long-term memory lasts **days**, **months**, **years**, or a **lifetime**.
- Storage capacity
  - Short-term memory holds an average of **~seven(?)** chunks.
  - Long-term memory has an essentially **unlimited** storage capacity.

Short-Term Memory



- Short-term memory lies between sensory and long-term memory, retaining information for about **5 seconds to 1 minute**.
- Retention is brief and capacity is limited, typically to 7 ± 2 items.
- Its contents are available to **conscious report**. (清醒的报告)
- Rehearsal can transfer information to long-term memory. (演练)

The Magical Number: 7 ± 2



George Miller
1920 -- 2012

- George Miller (1956) studied short-term memory capacity quantitatively.
- Young adults have a memory span of about **seven** units, or chunks, such as digits, letters, or words.
- Later research found that capacity depends on the type of chunk.
  - Span is about **seven** digits, **six** letters, or **five** words.
  - Longer words yield a smaller span than shorter words.
- Cowan (2001) later proposed a capacity of **four** chunks in young adults.

The Magical Number: 7 or 4?

- Cowan (2001) argued for a much smaller short-term memory capacity, challenging the traditional 7 ± 2 estimate.
- This capacity primarily reflects the focus of attention. It emerges under specific conditions, such as when *chunking* or *rehearsal* is prevented. (演练)
  - Active attention differs from passive storage.
- **Chunking**, such as grouping digits in a phone number, expands the amount remembered. The underlying capacity remains about four chunks.

  (分块记忆)

Working Memory

- Baddeley and Hitch proposed working memory based on experiments that simulated short-term memory impairment (损伤).
- Working memory is a *limited-capacity* system for temporarily storing and processing information. It supports many **complex** cognitive activities.
- It extends the A&S model.

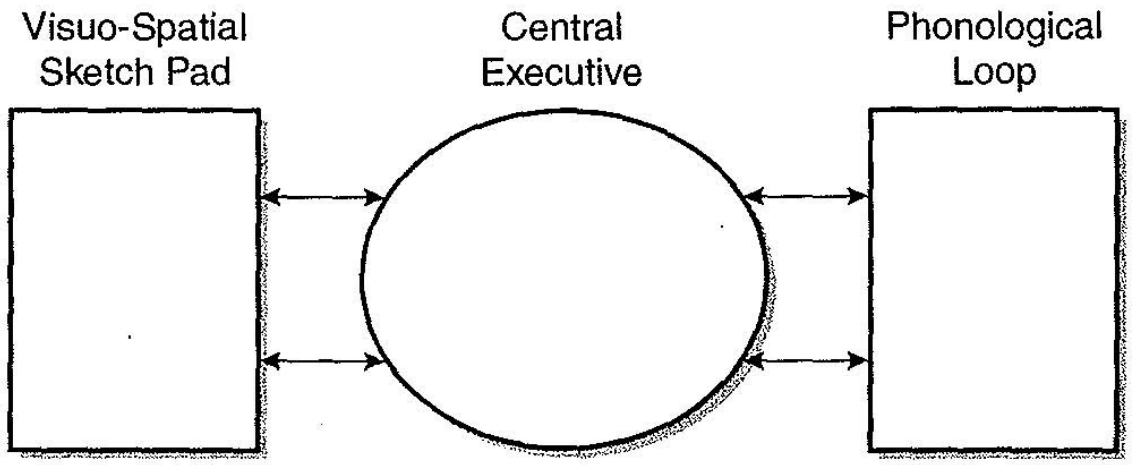
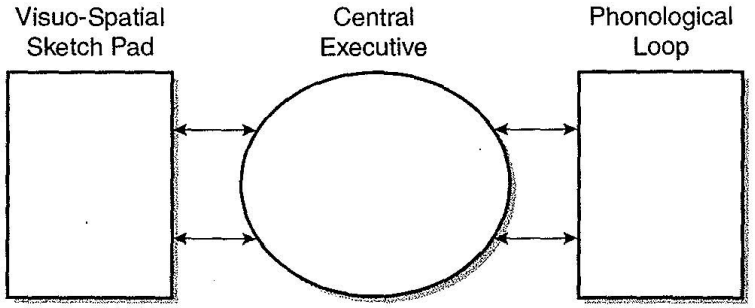


Figure from Baddeley (1986)
Source: https://en.wikipedia.org/wiki/Baddeley%27s_model_of_working_memory

Working Memory Components



中央执行

- **Central executive**: supervises the system and controls information flow between its subordinate components. (监控信息流)

语音循环

- **Phonological loop**: processes sound and speech information and stores verbal content. (处理语音信息)

视觉-空间速写板

- **Visuospatial sketchpad**: processes visual and spatial information. It can operate alongside the phonological loop with *little mutual interference*.
- The revised model (2000) adds an **episodic buffer**. It binds information across domains into integrated sequences of visual, spatial, and verbal content, such as memories of a story or a film scene.

情景缓冲区

Evidence for Working Memory

- Evidence from dual-task experiments (双任务实验)
  - People can perform two tasks in different domains, such as visual and verbal tasks, *almost as efficiently as each task alone*.
  - Performing two tasks in the *same domain* reduces performance compared with doing each task alone.
  - Example: reciting poetry while riding a horse is easier than reciting poetry while writing an essay.
- Why a central executive? In Alzheimer's disease, its decline can leave single-task performance intact while making two or more simultaneous actions difficult.

(阿尔兹海默症)

Source: https://en.wikipedia.org/wiki/Baddeley%27s_model_of_working_memory

Overview

- Memory Systems
- Short-Term Memory
- Working Memory
- **Memory Mechanisms in AI/LLMs**

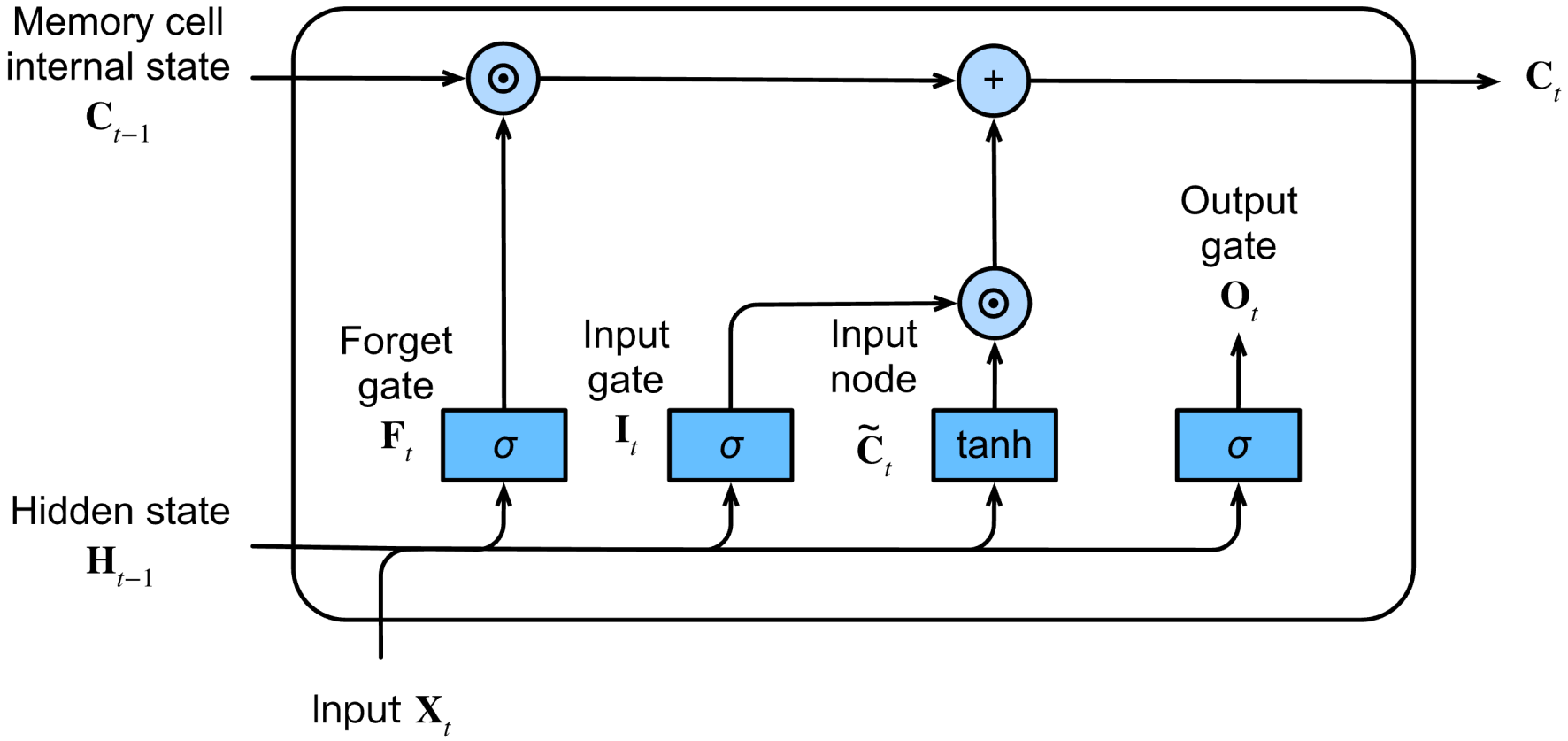
Artificial Intelligence and Memory

- Intelligence or memory?
- Early AI research sought scalability and tried to minimize the role of memorization in task performance.
  - Example: training, validation, and test splits help prevent overfitting.
- More recently, AI has pursued specificity by building agents that use memory to meet individual human needs.
  - Example: retaining specific information across long contexts.

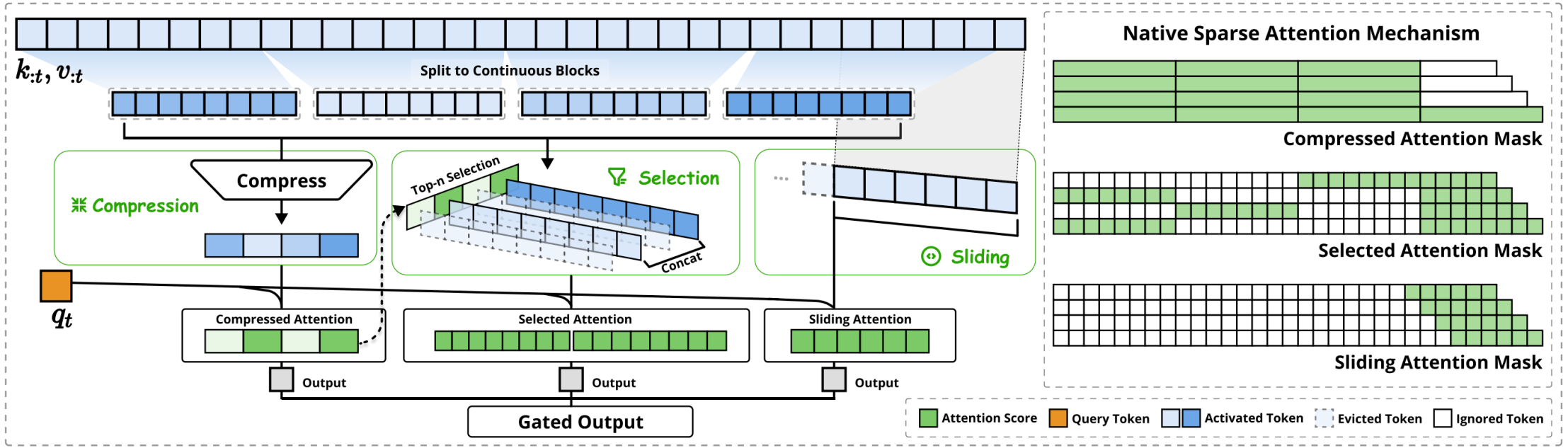
Source: https://d2l.ai/chapter_recurrent-modern/lstm.html

Simple “memory” mechanism in LM

- Long Short-Term Memory (LSTM)



LLM: Everything in context window ⇒ Memory



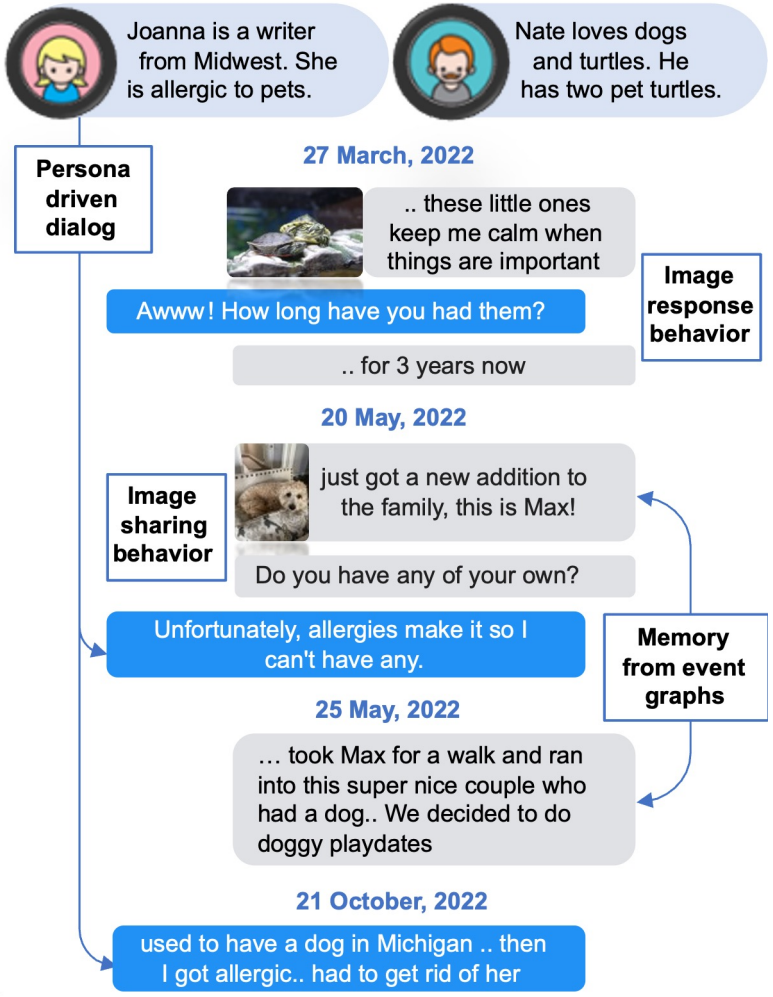
q_t: query vector for the token at position t. $k_{:t}, v_{:t}$: keys and values for previous tokens.

Source: Yuan et al. (2025), Native Sparse Attention, Fig. 2. https://arxiv.org/abs/2502.11089

Long-Term Conversation

Source: https://snap-research.github.io/locomo/

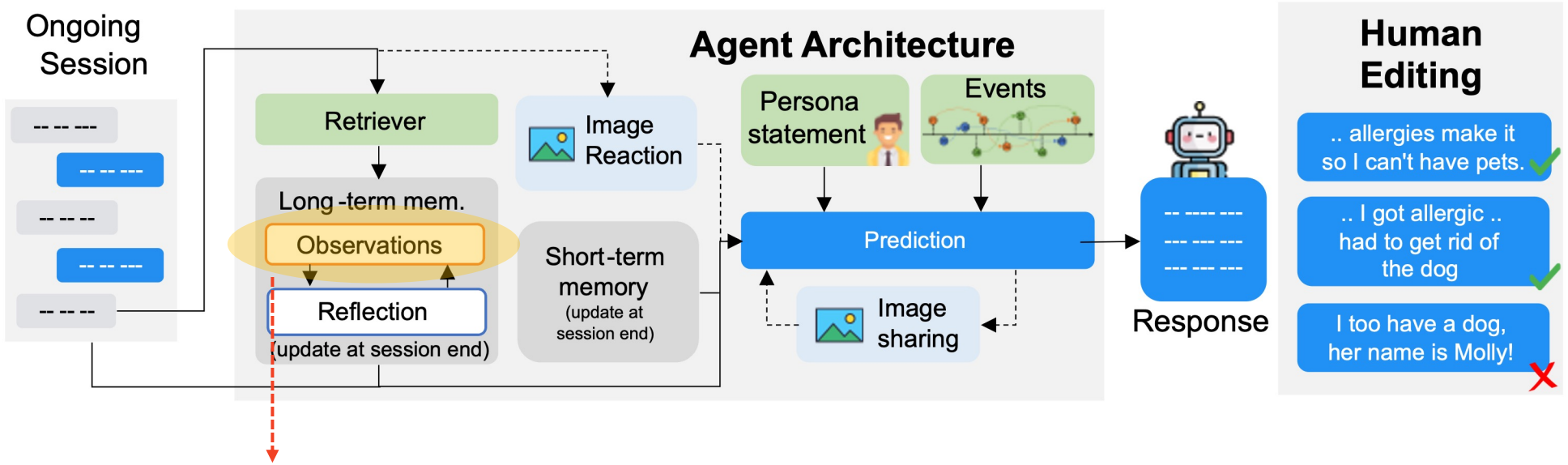
- **LoCoMo**: A dataset of very long-term conversations, each encompassing **300 turns** and **9K** tokens on avg., over up to 35 sessions.
- Example: Two agents, each initialized with an LLM. Unique **persona statements** are assigned to each agent.
- Create a **temporal event graph** for each agent to mirror real-life experiences
- **Agent architecture** is utilized for each agent, enabling them to effectively memorize and reflect conversation history in ongoing dialogues.



Agent architecture

Source: https://snap-research.github.io/locomo/

- The agent is equipped with an **external memory** and **reflection** module to retrieve relevant history for dialog generation
- Persona statement and a timeline of causally connected events in their life.

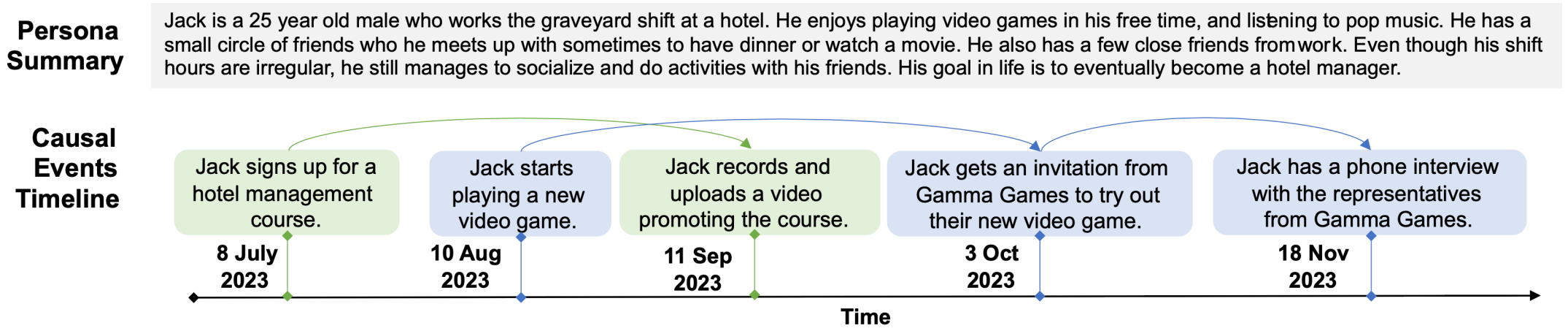


a database of assertions (observations) about each speaker's life and persona

Temporal Event Graph

Source: https://snap-research.github.io/locomo/

- Each event is generated in accordance with the specified persona and causal connections between events.



Evaluation

Source: https://snap-research.github.io/locomo/

- **Question Answering**. Agents need to "recall" past context correctly to integrate relevant information into future responses.
  - Single-hop, multi-hop, common sense & world know., adversarial, temporal
- **Event Graph Summarization.** Agents also need to recognize long-range causal (长程因果) and temporal connections (时间联系) in the dialogues to generate empathetic and relevant responses.
- **Multi-modal Dialog Generation.** Conversational agents need to utilize relevant context recalled from past conversations to generate responses that are consistent with the ongoing narrative.

Findings

Source: https://snap-research.github.io/locomo/

- **Long-context LLMs** and **RAG** demonstrate effectiveness in **QA** tasks: improving "memory" capabilities of LLMs, but still significantly lag behind human levels (by 56%), especially in temporal reasoning, (by 73%);

Category	Model	Context Length	Answer Prediction (F1)					
			Single Hop	Multi Hop	Temporal	Open Domain	Adversarial	**Overall**
Human	Human	-	95.1	85.8	92.6	75.4	89.4	87.9
Base	Mistral-Instruct-7B	8K	10.2	12.8	16.1	19.5	17.0	13.9
	Llama-2-Chat-70B	4,096	19.7	14.4	13.3	15.9	22.1	17.9
	GPT-3.5-turbo	4,096	**29.9**	23.3	**17.5**	**29.5**	12.8	22.4
	GPT-4-turbo	4,096	23.4	**23.4**	10.4	24.6	**70.2**	**32.1**
Long context	GPT-3.5-turbo-16K	4K	31.7	25.4	16.8	27.6	**13.1**	24.1
		8K	38.8	31.2	21.0	35.0	8.4	25.2
		12K	51.1	40.4	**25.0**	36.5	6.4	33.5
		16K	**56.4**	**42.0**	20.3	**37.2**	2.1	**37.8**

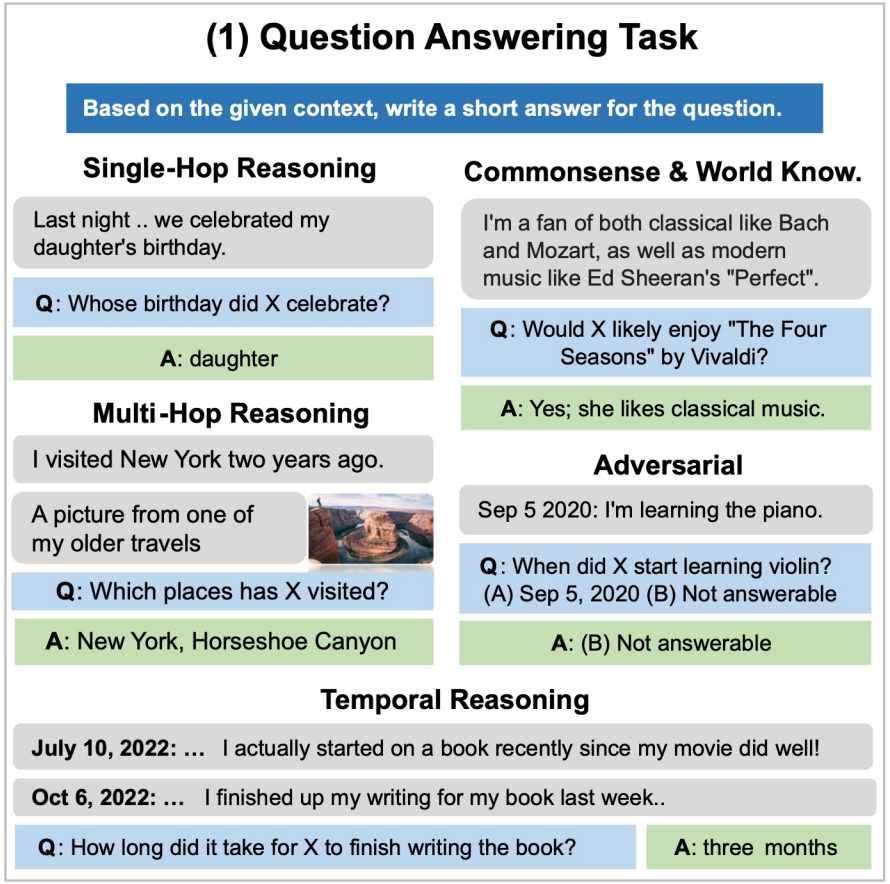
Table 2: **Question answering performance** of Base and Long-context models. Optimal performance is in **bold**. Results are based on F1-score for answer prediction; higher is better.

Findings

Source: https://snap-research.github.io/locomo/

(对抗性问题)

- **Long-context LLMs has significant hallucinations:** difficulty with *adversarial* questions in QA and event graph summarization task.



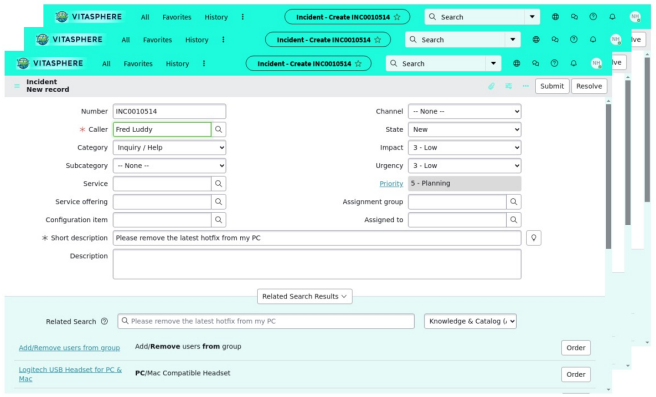




LongMemEval-V2 (2026)

Can a web agent learn from experience from customized web environment?

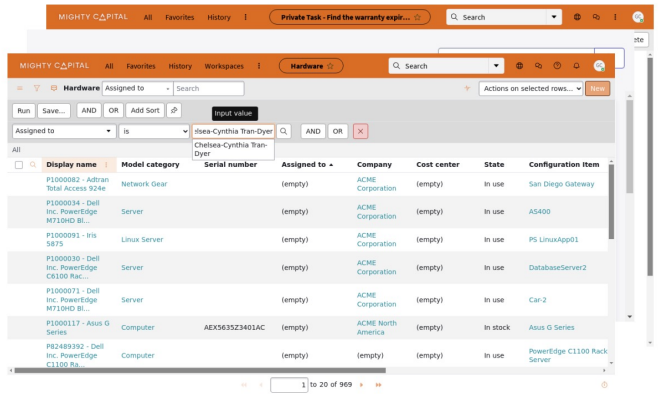
451 questions based on histories of up to 500 trajectories / 115M tokens.



🎯 **Task**: Create an Incident

🛠️ **Actions**: Fill → Fill → Select → Click …

✅ **Result**: Success



🎯 **Task**: Find Hardware Expiration Date

🛠️ **Actions**: Click → Fill → Tab_Focus → …

✅ **Result**: Success

Source: Wu et al. (2026), LongMemEval-V2. https://arxiv.org/abs/2605.12493

LongMemEval-V2 (2026)

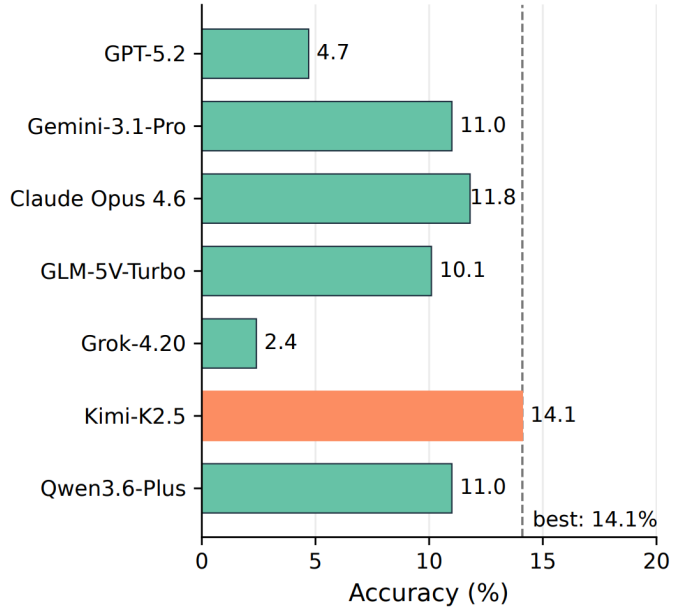
- Test AI-agent's capability as a colleague.
- Question: What does an experienced colleague **internalize** after repeatedly working in an environment?

Five abilities

- Static state recall（静态回忆）
- Dynamic state tracking（动态追踪）
- Workflow knowledge
- Environment gotchas（环境陷阱）
- Premise awareness（前提意识）

Context	Overall	Static	Dynamic	Workflow	Gotchas
Qwen3.5-9B (thinking enabled)					
No context	0.016	0.000	0.015	0.155	0.136
Oracle trajectories	0.596	0.566	0.668	0.718	0.310
Oracle slices + notes	0.825	0.908	0.879	0.750	0.484
GPT-5.4-mini (medium reasoning)					
No context	0.045	0.025	0.010	0.075	0.171
Oracle trajectories	0.653	0.660	0.696	0.697	0.484
Oracle slices + notes	0.863	0.950	0.843	**0.905**	0.467
Codex + GPT-5.4-mini (xhigh reasoning)					
Oracle trajectory files	**0.897**	**0.986**	**0.947**	0.815	**0.517**

Up: Coding agent harness enabled with trajectory; Below: LLM performance without trajectory history

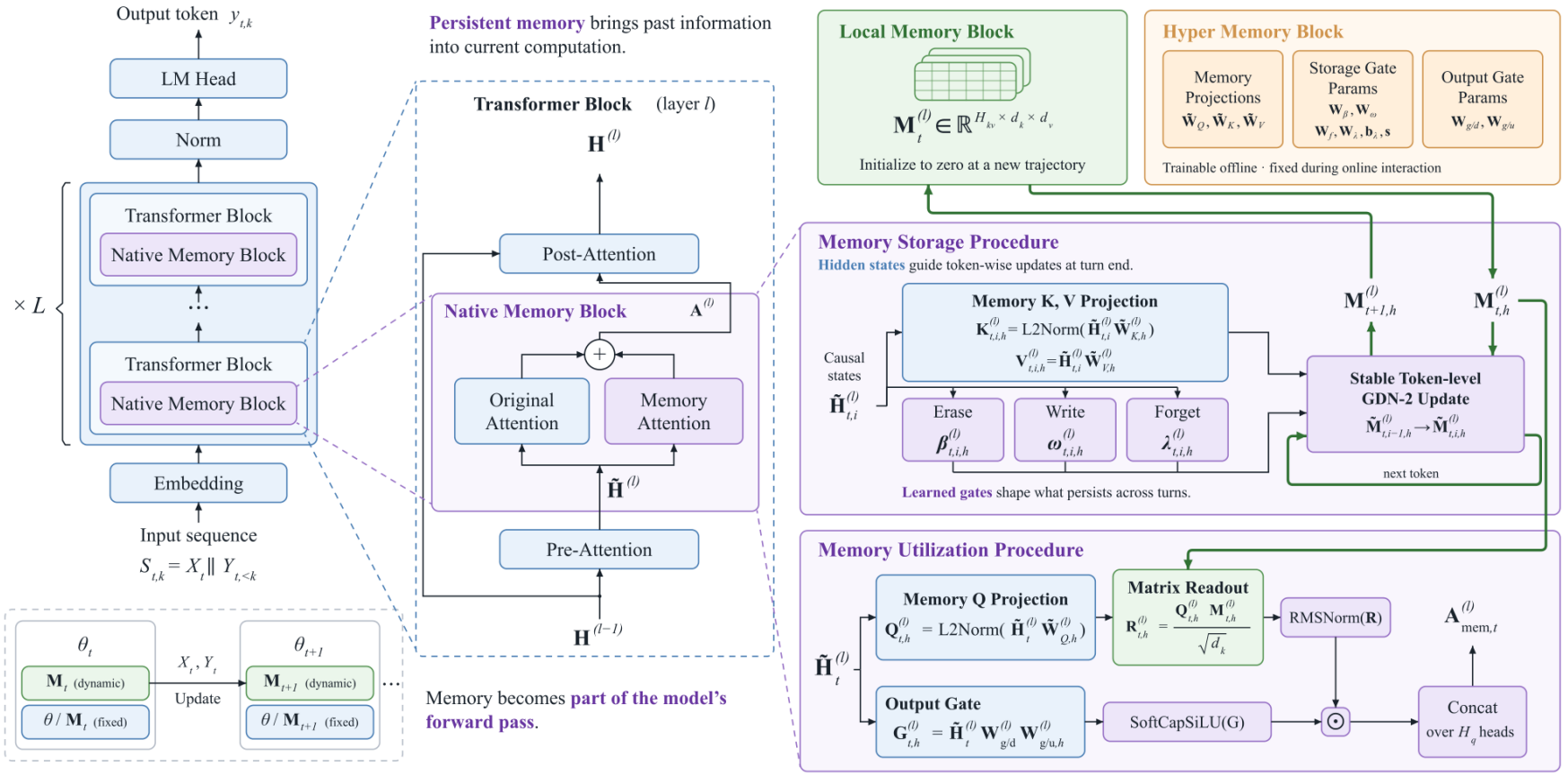


Source: Wu et al. (2026), LongMemEval-V2. https://arxiv.org/abs/2605.12493

MemT-1.0: Native Memory (2026)

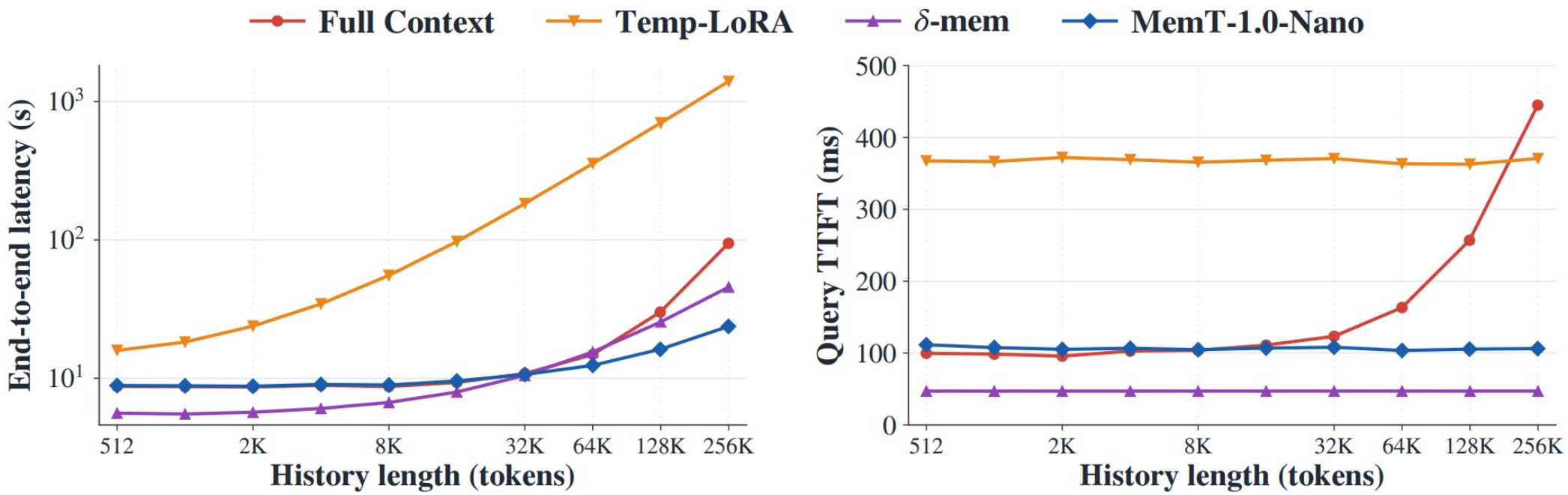
(原生/参数化记忆)

Parametric memory: Dynamic local memory that is learned and persists across turns.



Source: MemTensor Team (2026), MemT-1.0, Fig. 4. https://arxiv.org/abs/2610.26825

Parametric Memory: Advantages



Low latency and memory cost.

Source: MemTensor Team (2026), MemT-1.0, Fig. 4. https://arxiv.org/abs/2610.26825

Gap to External Memory System

Method	Backbone	< 16K			16–32K			32–64K		> 64K			All
		MB	IM	PM	PSB	LCM	PMS	MMR	MO	LME	MAB	BEAM	AVG
Textual Memory													
MemOS	DS	78.4	**68.2**	67.1	92.9	**85.7**	45.9	**55.5**	77.0	**83.2**	27.8	**72.6**	**68.6**
	QW	79.9	61.1	**67.6**	97.3	83.4	54.7	52.6	82.6	80.8	21.8	71.3	68.5
Mem0	DS	75.8	45.9	67.2	89.4	79.0	**66.0**	47.5	75.7	72.6	30.3	71.4	65.5
	QW	78.5	60.4	63.2	97.3	80.9	54.9	41.4	79.4	78.9	20.6	70.3	66.0
Parametric Memory													
MemT-1.0-Flash	–	**73.5**	**49.2**	**61.6**	97.5	**65.1**	33.5	20.5	**64.1**	**37.2**	**17.4**	**37.0**	**50.6**
MemT-1.0-Nano	–	71.8	43.1	52.3	95.7	57.5	**38.7**	**23.9**	49.9	32.6	15.5	35.8	47.0

Still a gap
But
promising

Source: MemTensor Team (2026), MemT-1.0, Fig. 4. https://arxiv.org/abs/2610.26825

Recap

- Memory Systems
- Short-Term Memory
- Working Memory
- Memory Mechanisms in AI/LLMs